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Paleozoic Paleomagnetism of Almaden, Spain: A Cautionary Note

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BACKGROUND, GEOLOGY, AND SAMPLING

The mercury mining district of Almaden occurs in an east-west trending syncline with stratified rocks ranging in age from Ordovician through Devonian (Figure 1). The area has been extensively mapped and described in the literature [e.g. *Bouyx*; 1970, *Bouyx and Saupe*, 1966; *Maucher and Saupe*, 1967; *Saupe*, 1967, 1971; *Tamain*, 1972; *Verges*, 1983, *etc.*]. A previous paleomagnetic study [*Perroud et al.*, 1991] has revealed two ancient components of magnetization (Table 1), with ENE and upward and ESE and downward directions, respectively. The ENE and upward component has been observed in sites at the southern flank of the syncline as well as at the eastern end of the syncline where the fold axis plunges westward; it passes the fold test, although the tilt-corrected directions are still fairly scattered ($\alpha 95 = 14$). The ESE and downward directions fail the fold test and represent a pervasive overprint in the district. *Perroud et al.* [1991] interpreted the ENE and upward component as primary and of reversed polarity. The choice of polarity, however, is not altogether clear and depends on estimates of the relative rotations the sampling area has undergone with respect to other parts of the Iberian Peninsula and the rest of Hercynian Europe. The reasoning of Perroud and colleagues went as follows: (1) the ESE and downward overprint is most likely of Carboniferous (Hercynian) age; (2) similar overprints in Portugal, northern Spain (e.g., Asturias) and in the Armorican Massif of France have similar downward shallow inclinations, whereas the declinations are SSW in Portugal and are SSW in France; (3) a counterclockwise rotation of the Iberian Peninsula associated with the Cretaceous opening of the Bay of Biscay can be corrected for by adding 35° to the declinations; (4) earlier rotations, related to the tightening of the Ibero-Armorican arc [*Perroud*, 1986; *Bachtadse and Van der Voo*, 1986] are most likely Late Carboniferous in age and must also be corrected for; (5) applying these corrections, the declinations for the Almaden overprint become SSW and aligned with those for the rest of Europe, whereas the declination of the "primary" component becomes southeasterly. Such a southeasterly declination is more likely to be reversed, according to Perroud and colleagues, and this interpretation leads to a northerly subtropical paleolatitude of 20 ° based on the observed inclination of -38°. Combining this paleolatitude with those from other Paleozoic Iberian rocks and with those expected for the northwest African margin of Gondwana, a paleolatitudinal drift history was obtained (Figure 2) which shows that Iberia and Gondwana may well have drifted northward together from Ordovician through Silurian times. In contrast, Gondwana moved southward in the Devonian away from Iberia, which presumably by now was already attached to, or in the vicinity of, the combined North America- Baltic Shield-Russian Platform or Old Red Continent [e.g., *Ziegler*, 1988]. This imaginative interpretation illustrated by Figure 2 hinges critically on the assumed primary age of the ENE and upward remanence of the Silurian Almaden volcanics. It is with respect to this last interpretation (and not with respect to the directional analysis of the remanence) that we feel our new observations are relevant. These observations force us to question the primary nature of the ENE and upward component.

Our sampling was designed to complement, not to duplicate, that of Perroud and colleagues; the locations of our four sites are shown in Figure 1. Three sites were collected in south dipping Ordovician flows on the northern flank of the syncline, whereas at one site in the center of the syncline we have sampled dolerites intrusive into Late Devonian strata, whose ages are well constrained by fossils [*Verges*, 1983]. The structural attitudes of the Ordovician sites could be well determined from sedimentary rocks, whereas the dolerites occur right at the center of the syncline where the fold axis orientation has changed from the westerly plunge that can be observed at the easternmost end of the fold to a 16 ° eastward plunge near or west of our site according to *Vergès* [1983, p. 228].

NEW PALEOMAGNETIC RESULTS

Natural Remanent Magnetization (NRM) intensities of the dolerites range from 9×10^{-5} to 2×10^{-2} A/m and for the Ordovician flows from 1×10^{-4} to 3×10^{-2} A/m. Thermal demagnetization analysis of the Devonian dolerites (Figures 3*a*-3*c*) reveals three components of magnetization: (1) a low-blocking temperature northerly and downward component A, removed in temperatures up to 250° or 300° C, (2) a southeasterly shallow downward component B removed between 250° and 500° C, and (3) an ENE and upward component C removed between 500° and 550° to 600° C. Component A we interpret as of recent origin, whereas component B (Figure 4*a*) is the same as the pervasive overprint observed by *Perroud et al.* [1991]. The ENE declinations of component C are well defined, whereas the inclination determinations, as seen in the vertical projections of Figures 3*a*-3*c*, are affected by somewhat "noisy" behavior of the samples during the higher demagnetization temperatures. We also note that the trajectories do not trend precisely to the origin, although this effect is not systematic. Clearly, these dolerites are less stable magnetically in high temperatures due to mineralogical changes in the samples during heating. Be that as it may, there is no doubt about the upward nature of the inclinations, and the

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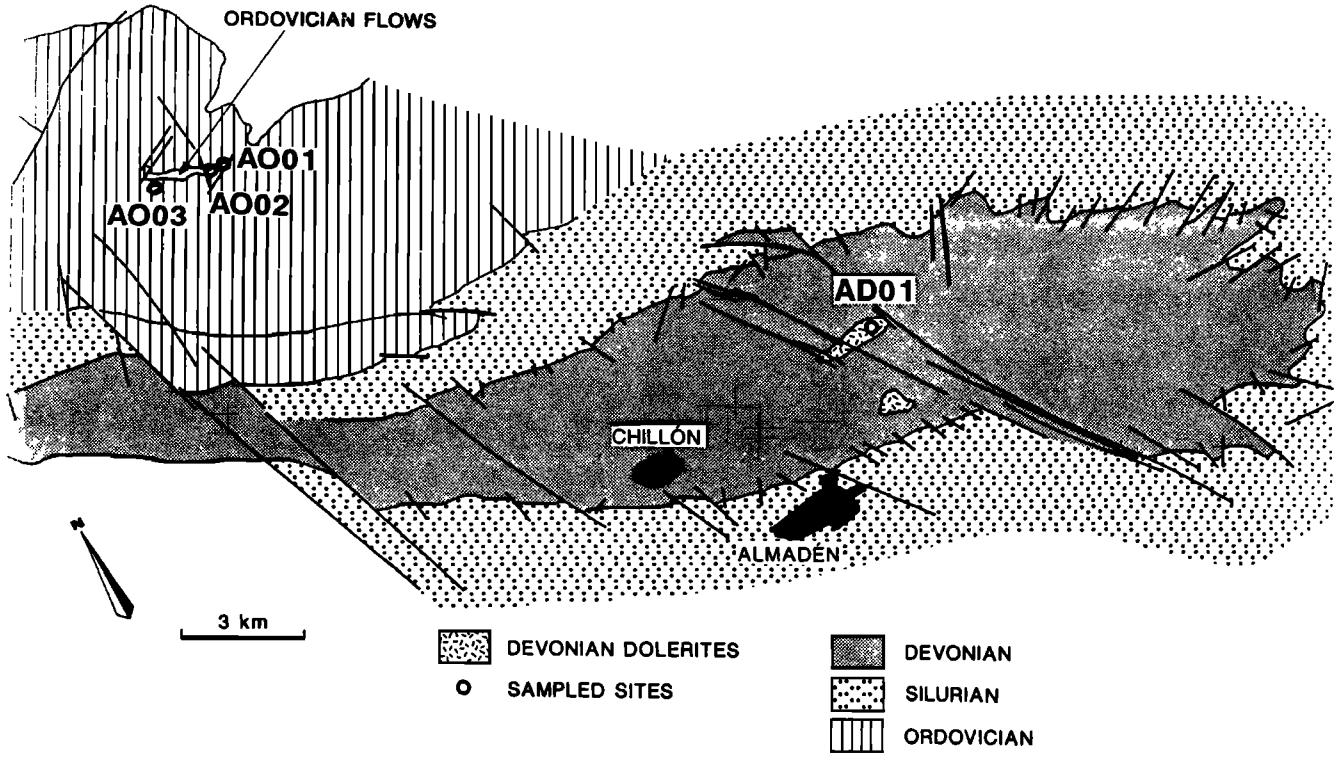


Fig. 1. Schematic geological map of the outcrop area with sampling sites. Map is based on *Vergés* [1983] and *García Sansegundo et al.* [1987].

directions obtained from principal component analysis [*Kirschvink*, 1980] for the dolerites are well grouped (Figure 4*b*). The mean direction without tilt correction (Table 1) is declination; inclination = 81/-37. If a correction is applied for the minor eastward plunge of the fold axis mentioned earlier, the inclination becomes more steeply upward. It is not clear whether this correction is necessary for our location; we note that our C direction and that of *Perroud et al.* [1991] are in good agreement with or without this correction (Figure 4*b*), given the large cone of confidence of the earlier result.

Thermal demagnetization of the Ordovician flow samples reveals mostly the ESE and downward overprint or B component (Figure 4*a*), but a few samples show shallow to upward directions in the northeast quadrant (Figures 3*d*-3*f*). However, the number of samples showing this C component direction and the sometimes noisy demagnetization behavior do not allow us to calculate a meaningful average direction; we merely note its existence.

TABLE 1. Paleomagnetic Results From Almaden

Sites	*n/ns*	Dec	Inc	α_{95}	References
Silurian Spillites					
Component I (11 sites)		62	-36	14	Perroud et al. [1991]
Component II (20 sites)		109	11	9	Perroud et al. [1991]
Devonian Dolerites					
AD01, component A	9/19	349	66	5	this study
AD01, component B	13/19	128	32	7	this study
AD01, component C	8/19	81	-37	10	this study
Ordovician Flows					
AO01, component B	8/12	105	32	19	this study
AO02, component B	13/14	148	36	13	this study
AO03, components B&C	10/28	56	14	21	this study

n/ns is the number of samples demagnetized/used in statistics; Dec is declination; Inc is inclination.

INTERPRETATION AND CONCLUSIONS

Our observations of the B component are in accord with those of *Perroud et al.* [1991] and strengthen their interpretation, although our declinations show a greater spread between southerly and easterly. On the other hand, the occurrence of the C component in the dolerites of Late Devonian or younger age, as well as the trace of its existence in the Ordovician samples, raises serious questions

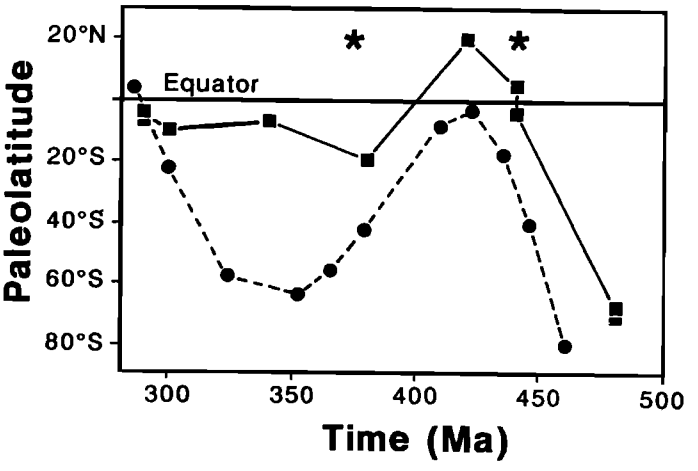


Fig. 2. Paleolatitude versus time curve for northern Spain (solid curve) and the northern margin of Africa (dashed curve) from Ordovician through Carboniferous time [after *Perroud et al.*, 1991]. The two stars represent the paleolatitudes of the results for the Ordovician and Devonian rocks of our study, on the (tenuous) assumption that the magnetization age is primary.

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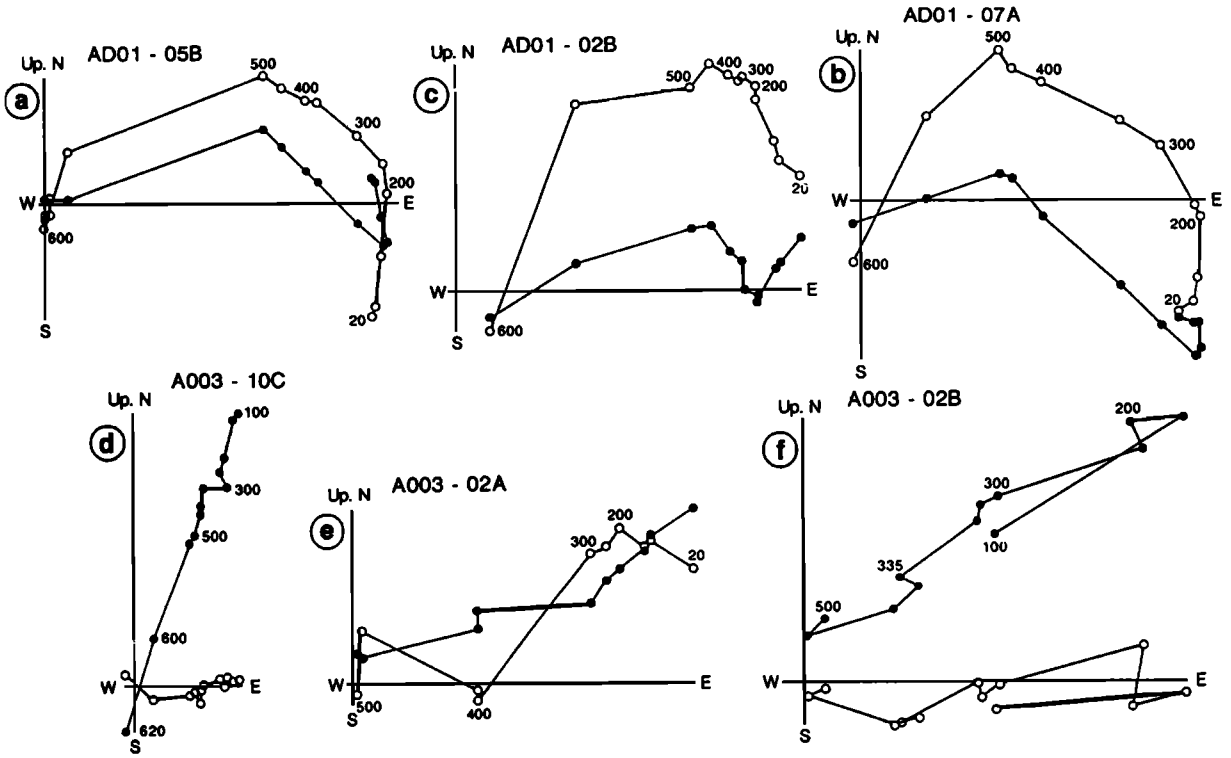


Fig. 3. Demagnetization diagrams in in situ coordinates of typical samples from (*a-c*) the Devonian dolerites and (*d-f*) the Ordovician flows. Closed (open) symbols represent projections onto the horizontal (vertical) plane. The dolerites yield three components of magnetization (A, B, and C in Table 1), whereas the Ordovician samples show a near-horizontal to shallowly upward component in the northeast quadrant.

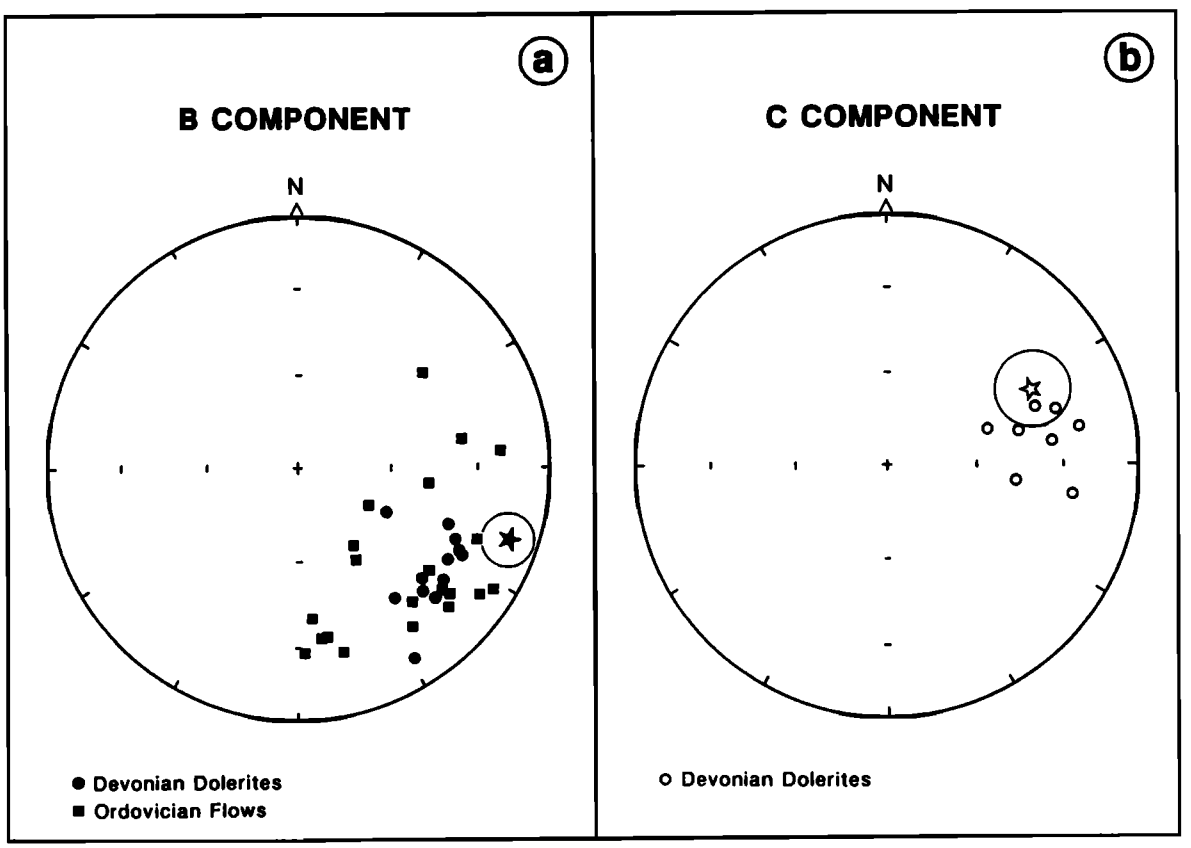


Fig. 4. Equal-area plots of sample directions of magnetization. Closed (open) symbols represent projections onto the lower (upper) hemisphere. The stars are the mean directions obtained by *Perroud et al.* [1991] plotted with their cones of confidence.

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about the interpretation of Perroud and colleagues that this magnetization is primary and of Silurian age. We have plotted the paleolatitudes, assuming a reversed polarity (following *Perroud et al.* for the time being) in Figure 2 as a pair of stars, and it can be seen that these additions destroy the simple coherence previously thought to exist between the drift patterns of Gondwana and Iberia. For all Atlantic-bordering continents and blocks (Laurentia, Baltica, Avalon, Southern Britain, the Asturian-Cantabrian area of northern Spain, the Armorican Massif), the observed geomagnetic field changes are large for the Ordovician through Devonian interval. We consider it therefore very unlikely that the ENE and upward or C directions are primary in the Ordovician and Silurian as well as the Devonian rocks of Almaden. The only logical conclusion is that these are remagnetizations, if not in all rocks, at least in those of Ordovician and Silurian age.

Questions then arise as to what age these magnetizations have, what structural corrections are or are not to be applied, what the polarity assignment (normal instead of reversed?) should be, and what these remanences mean for the Paleozoic drift history of the Almaden region. Unfortunately, there are too many uncertainties in the equation for it to be solved. We note, however, one intriguing coincidence, namely, between the results from Beja, southern Portugal [*Perroud et al.*, 1985; *Ruffet*, 1990] and those from Almaden. Both have ENE declinations and upward inclinations, although those from Beja are steeper. The age of the Beja Gabbro has now been determined by *Ruffet* [1990] as Early Carboniferous; their structural corrections are of necessity somewhat uncertain. The polarity assignment for the Beja rocks has traditionally been normal, indicating a southerly paleolatitude. Without belaboring the issue further, in view of the ambiguities, we suggest that the Almaden C component may also be of normal polarity and of Early Carboniferous age. If so, these results exist in a bit of vacuum as paleomagnetic observations for Hercynian Europe for this age are very scarce, but what seems clear is that the rotational and drift history for Iberia for this time may be much more complicated than previously thought.

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